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```julia
begin
    using Plots
    using PlutoUI
    using PlutoTeachingTools
    using LaTeXStrings
    using QuadGK
    using Printf
    using HypertextLiteral: @htl, @htl_str
end
```

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Numerical integration

Integration is an important operation in engineering and the physical sciences, but evaluating integrals analytically can be quite challenging. For such cases employing *numerical* integration techniques is a good alternative, which sometimes turns out to be no less accurate.

Let's start with an easy analytic case. The integral $\int_0^1 e^x dx$ can be computed rather elegantly by analytical means, since the anti-derivative of e^x is again e^x. Therefore

$$\int_0^1 e^x dx = [e^x]_0^1 = e^1 - 1 \approx 1.718281828459045.$$

```
exact = 1.718281828459045
```

```
exact = exp(1) - 1
```

To perform the integration numerically, we can employ the Julia package `QuadGK`, which implements an all-purpose numerical integration routine:

```
1.7182818284590453
```

```
let
    Q, error_estimate = quadgk(x -> exp(x), 0, 1)
    Q
end
```

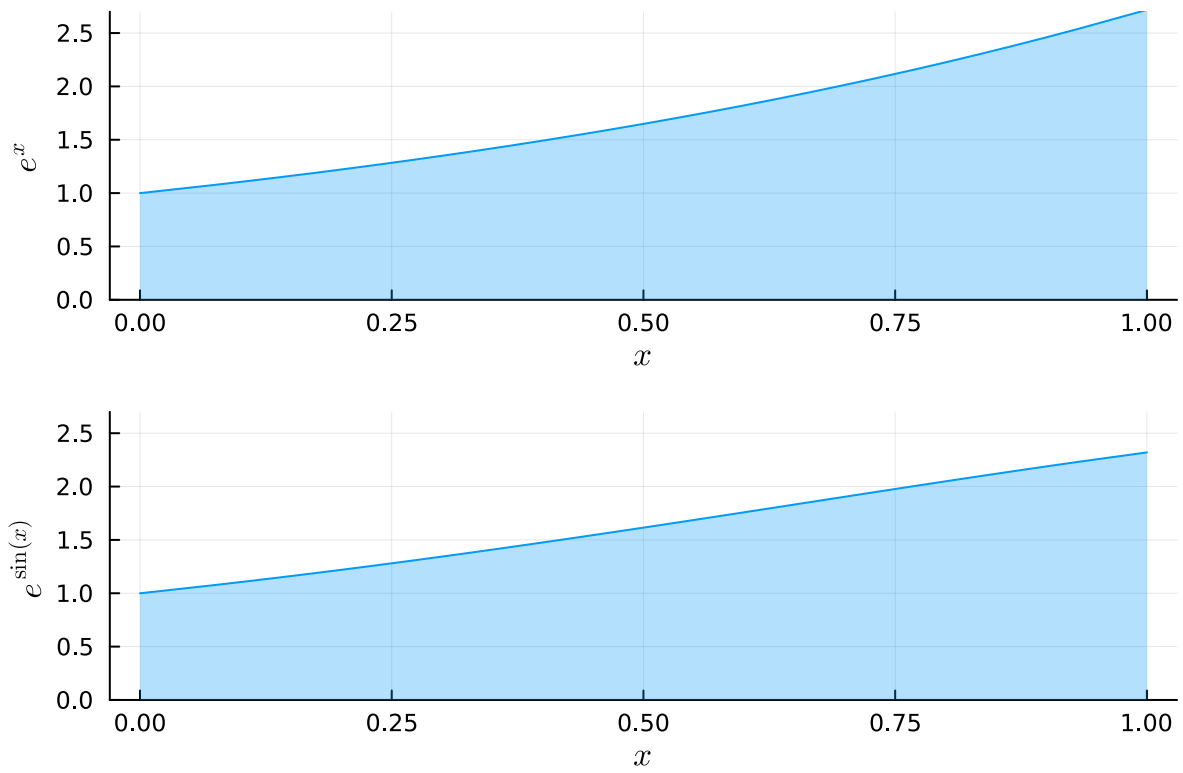
As we observe, the difference to the analytical result is tiny and on the order of the floating-point precision.

However, the numerical approach is much more flexible. For example $e^{\sin(x)}$ has no useful anti-derivative. Still the numerical approach works just as well to compute $\int_0^1 e^{\sin(x)} dx$:

```
1.6318696084180515
```

```
let
    Q, error_estimate = quadgk(x -> exp(sin(x)), 0, 1)
    Q
end
```

What is remarkable when looking at the graphs of these two functions is that they are very similar ... and for one case the area under the curve boils down to simple basic calculus and the other is impenetrable analytically. However, from a numerical standpoint they are basically the same problems.




```
begin
    p = plot(exp, 0, 1, fill=0, fillalpha=0.3, xlabel=L"x", ylabel=L"e^x", ylims=(0, 2.7), label="")
    q = plot(x -> exp(sin(x)), 0, 1, fill=0, fillalpha=0.3, xlabel=L"x", ylabel=L"e^{\sin(x)}", ylims=(0, 2.7), label="")

    plot(p, q, layout=(2, 1))
end
```

Simple and composite quadrature formulas

The task of **numerical integration** is to **approximate an integral** $I = \int_a^b f(x)\,dx$. We want to achieve this by **sampling the function** at N carefully selected points $t_i \in [a, b]$ with $i = 1, \ldots N$, followed by taking linear combinations of the results. We thus work towards approximations of the form

$$I = \int_a^b f(x)\,dx \approx \underbrace{\sum_{i=1}^{N} \alpha_i f(t_i)}_{=Q_a^b(f)}.$$

The expression on the right is usually called a **quadrature formula** $Q_a^b(f)$ while the sampling points t_i are called **quadrature nodes**. There are two common strategies to construct such formulas:

Simple quadrature

Approximate f by a function $\widetilde{f}$ that is easy to integrate, then perform $\int_a^b \widetilde{f}(x)\,dx$. This result is used as the approximation of the integral I. Typically, one takes $\widetilde{f}$ to be a polynomial approximation to f, since

- **polynomials can be integrated in closed form** and
- we know standard procedures to to approximate functions as polynomials, namely **polynomial interpolation**, which we discussed in the chapter on Interpolation.

Recall that to construct a p-th degree polynomial approximation $\widetilde{f}$ we require $p+1$ samples of the function f. Therefore a simple quadrature formulas employing N quadrature nodes effectively works by constructing an $(N-1)$-th degree polynomial interpolation of f on $[a,b]$.

Composite quadrature

Here we first subdivide the interval $[a,b]$ into a set of n subintervals $[\tau_{i-1},\tau_i]$ with $a=\tau_0<\tau_1<\cdots<\tau_{n-1}<\tau_n$. We therefore have $n+1$ integral boundaries. In general the τ_i can be distributed arbitrarily in the interval $[a,b]$. However, for simplicity we will assume **intervals of equal size** using the definition

$$\tau_i = a + i\,h, \qquad h = \frac{b-a}{n} \qquad \text{where } i = 0,1,\ldots,n. \tag{1}$$

in this class. Now notice that

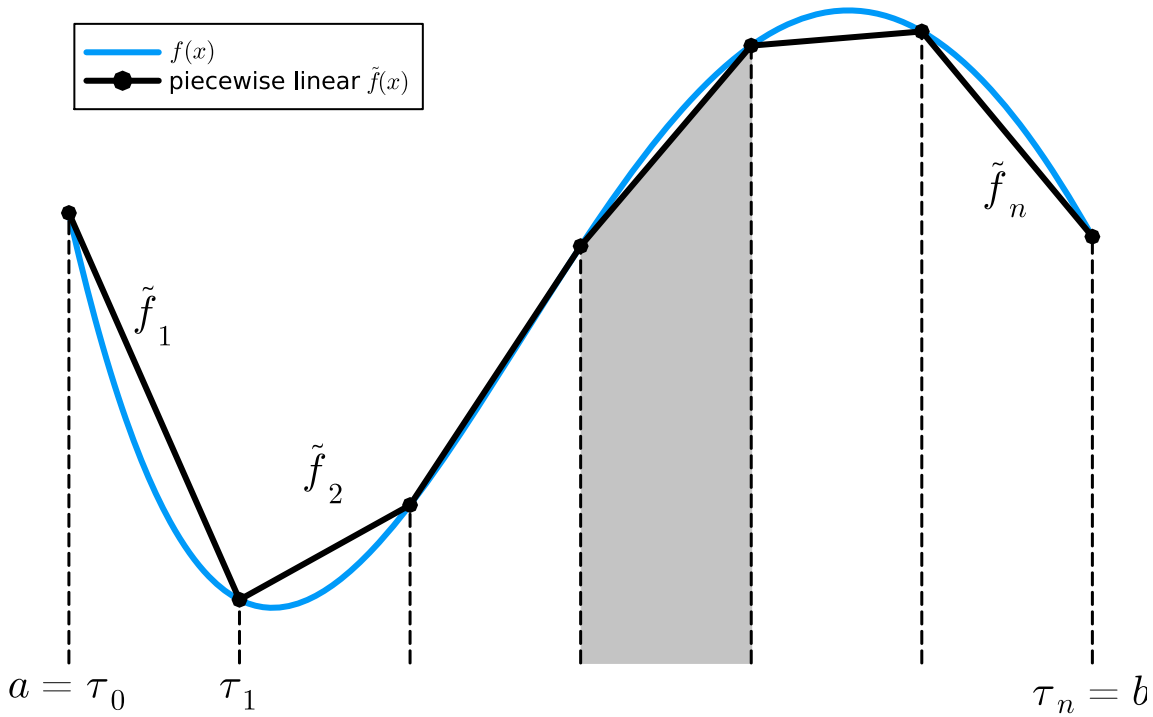
$$I = \int_a^b f(x)\,dx = \sum_{i=1}^{n} \int_{\tau_{i-1}}^{\tau_i} f(x)\,dx \approx \sum_{i=1}^{n} \int_{\tau_{i-1}}^{\tau_i} \widetilde{f}_i(x)\,dx \tag{2}$$

where in the last step we follow the same idea as in the *simple quadrature* discussion above, namely we approximate the function f on the interval $[\tau_{i-1},\tau_i]$ by a polynomial $\widetilde{f}_i$.

Notice that each approximation $\widetilde{f}_i$ is only defined on the subinterval $[\tau_{i-1},\tau_i]$. However, by stitching each of them together we also obtain a global approximation $\widetilde{f}$ to f, namely

$$\tilde{f}(x) = \begin{cases} \tilde{f}_1(x) & \text{if } x \in [t_0, t_1] \\ \tilde{f}_2(x) & \text{if } x \in [t_1, t_2] \\ \vdots & \vdots \\ \tilde{f}_n(x) & \text{if } x \in [t_{n-1}, t_n] \end{cases}$$

See the figure below, which illustrates this for the case of constructing linear approximations on each subinterval $[\tau_{i-1}, \tau_i]$. We recognise $\tilde{f}$ to be a **piecewise polynomial approximation** to f.



Observation: Composite quadrature rules

Composite quadrature rules are based on the idea of constructing a **piecewise polynomial approximation** and then integrating this approximation analytically instead of the original function.

Composite quadrature rules are the state of the art for numerical integration and provide in general more accurate results compared to simple quadrature rules.

▶ **Why prefer composite quadrature rules over simple ones ?**

Notational confusion N versus n versus h

In this class we will use the following notation:

- N **number of integration nodes** labelled $t_1, t_2, \ldots, t_N$
- n **number of subintervals** $[\tau_{i-1}, \tau_i]$, i.e. $n+1$ integral boundaries $\tau_0, \tau_1, \ldots, \tau_n$
- h **width of the subintervals** $h = \tau_i - \tau_{i-1}$

In general $N > n+1$ and therefore the *distance between the nodes* is usually smaller than h.

Trapezoidal rule

Our goal is to compute the integral $I = \int_a^b f(x)\,dx$, that is the area under the blue curve in the figure above.

Following our discussion about composite quadrature rules a first idea is to perform **linear interpolation** in each of the subintervals $[\tau_{i-1}, \tau_i]$ with $i = 0, \ldots, n$. In other words we aim to approximate in each interval $[\tau_{i-1}, \tau_i]$ the function f by a first degree polynomial $\widetilde{f}_i$, i.e. a straight line. This leads to the **black line segments** in the above figure.

To fit such a linear polynomial $\widetilde{f}_{1,i}$ two nodal points are sufficient in the subinterval. Knowing f on the interval boundaries, i.e. having access to the data points $(\tau_{i-1}, f(\tau_{i-1}))$ and $(\tau_i, f(\tau_i))$ is thus sufficient. We construct an interpolynomial using the based on the Lagrange basis described previously. Using the aforementioned two datapoints we thus obtain

$$\widetilde{f}_{1,i}(x) = f(\tau_{i-1}) \underbrace{\frac{x-\tau_i}{\tau_{i-1}-\tau_i}}_{\text{Lagrange function of } \tau_{i-1}} + f(\tau_i) \underbrace{\frac{x-\tau_{i-1}}{\tau_i-\tau_{i-1}}}_{\text{Lagrange function of } \tau_i}.$$

Recalling the definition of the subintervals (1), i.e. $\tau_i = a + i\,h$, this is equal to

$$\widetilde{f}_{1,i}(x) = -\frac{f(\tau_{i-1})}{h}(x-\tau_i) + \frac{f(\tau_i)}{h}(x-\tau_{i-1}).$$

Finally we follow equation (2) and integrate these line segments instead of the blue curve to obtain an approximation for our desired integral:

$$
\begin{aligned}
I = \int_a^b f(x)\,dx &= \sum_{i=1}^{n} \int_{\tau_{i-1}}^{\tau_i} f(x)\,dx \\
&\approx \sum_{i=1}^{n} \int_{\tau_{i-1}}^{\tau_i} \widetilde{f}_{1,i}(x)\,dx \\
&= \sum_{i=1}^{n} \left(\frac{f(\tau_{i-1})}{h} \int_{\tau_{i-1}}^{\tau_i} \tau_i - x\,dx + \frac{f(\tau_i)}{h} \int_{\tau_{i-1}}^{\tau_i} x - \tau_{i-1}\,dx \right) \\
&= \sum_{i=1}^{n} \left(\frac{f(\tau_{i-1})}{h} \frac{h^2}{2} + \frac{f(\tau_i)}{h} \frac{h^2}{2} \right)
\end{aligned}
$$

where we used that both $\int_{\tau_{i-1}}^{\tau_i} \tau_i - x\,dx$ and $\int_{\tau_{i-1}}^{\tau_i} x - \tau_{i-1}\,dx$ describe both the area of a triangle with width h and height h, thus have a value of $\frac{h^2}{2}$.

Rearranging this sum differently leads to the final formula

$$
\begin{aligned}
I = \int_a^b f(x)\,dx \approx T_n(f) &:= \frac{h}{2} f(\tau_0) + h f(\tau_1) + \cdots + h f(\tau_{n-1}) + \frac{h}{2} f(\tau_n) \\
&= \frac{h}{2} f(\tau_0) + \sum_{i=1}^{n-1} h\, f(\tau_i) + \frac{h}{2} f(\tau_n).
\end{aligned}
$$

This formula is called the **trapezoidal rule**. This name stems from a more geometric way of constructing the trapezoidal formula: as can be captured in the visualisation above, one can think of the formula as summing the areas of the trapezoids defined by the quadrature nodes (see shaded grey).

▸ **Alternative derivation using our results on piecewise polynomial interpolation**

The trapezoidal rule is just one representative of the wide class of numerical integration formulas. A general definition is:

Definition: Numerical integration formula

A **numerical integration formula** for the N equispaced **quadrature nodes** t_i, $i = 1, \ldots, N$ is the set of **weights** $w_1, w_1, \ldots w_N$, such that for an integrand $f : [a, b] \to \mathbb{R}$

$$
\int_a^b f(x)dx \approx Q_a^b(f) = \frac{b-a}{N-1} \sum_{i=1}^{N} w_i f(t_i). \tag{3}
$$

The weights w_i are independent of f.

An older and still frequently used name for numerical integration is **quadrature**.

By comparing with our derivation above, we realise:

Observation: Trapezoid formula

The trapezoid formula is the numerical integration of the form (3) where the N nodes are just equal to the $n+1$ subinterval boundaries, i.e. we have $N = n+1$ and

$$\begin{aligned} t_1 &= \tau_0 = a \\ t_2 &= \tau_1 = a + 1h \\ t_3 &= \tau_2 = a + 2h \\ &\vdots \\ t_{n+1} &= \tau_n = b. \end{aligned}$$

Here we used the subinterval width $h = \frac{b-a}{n} = \frac{b-a}{N-1}$. The associated weights are

$$w_i = \begin{cases} 1 & \text{if } 0 < i < N \\ \frac{1}{2} & \text{if } i = 0 \text{ or } i = N \end{cases}$$

We will see later that nodes and interval boundaries do not agree for more general formulas.

An implementation of the trapezoidal rule is:

```
trapezoid (generic function with 1 method)
```

```julia
function trapezoid(f, a, b, n)
    # f:  Function
    # [a, b]: Interval to integrate over
    # n:  Number of subintervals
    h = (b - a) / n
    τ = range(a, b, length=n+1)  # Interval boundaries
    y = [f(τₙ) for τₙ in τ]
    integral = h * (0.5y[1] + sum(y[2:n]) + 0.5y[n+1])
    (; integral, h)
end
```

Let's test this formula numerically on the simple integral from earlier, i.e.

```
f (generic function with 1 method)
```

```julia
f(x) = exp(x)
```

and

$$I = \int_0^1 e^x \, dx = e - 1$$

which is approximately 1.7182818.

We consider a sequence of results where we double the number of integration points:

```
ns = ▸[2, 4, 8, 16, 32, 64, 128, 256, 512, 1024, 2048]
```

```
ns = [2^i for i in 1:11]
```

The corresponding approximate integrals using the trapezoidal formula $T_n(f)$ and quadrature node distances are:

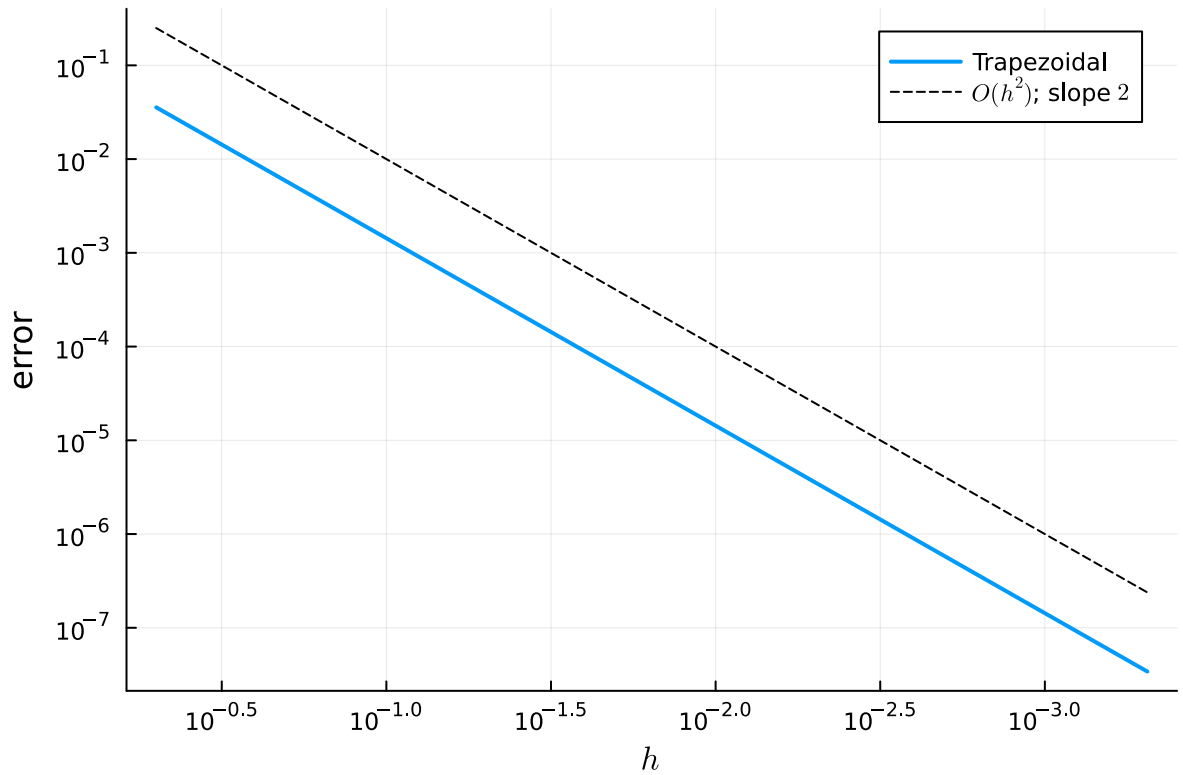
```
begin
    Tfs    = Float64[]
    hs     = Float64[]
    errors = Float64[]
    for n in ns
        res   = trapezoid(f, 0, 1, n)
        Tf    = res.integral  # Trapezoidal approximation
        h     = res.h         # Quadrature node distances
        error = abs(Tf - exact)
        @printf "n =%4d   T(f) = %.10f   error = %.10f\n" n Tf error

        push!(Tfs, Tf)
        push!(hs, h)
        push!(errors, error)
    end
end
```

```
n =   2   T(f) = 1.7539310925   error = 0.0356492640
n =   4   T(f) = 1.7272219046   error = 0.0089400761
n =   8   T(f) = 1.7205185922   error = 0.0022367637
n =  16   T(f) = 1.7188411286   error = 0.0005593001
n =  32   T(f) = 1.7184216603   error = 0.0001398319
n =  64   T(f) = 1.7183167869   error = 0.0000349584
n = 128   T(f) = 1.7182905681   error = 0.0000087396
n = 256   T(f) = 1.7182840134   error = 0.0000021849
n = 512   T(f) = 1.7182823747   error = 0.0000005462
n =1024   T(f) = 1.7182819650   error = 0.0000001366
n =2048   T(f) = 1.7182818626   error = 0.0000000341
```

We notice that the error decreases roughly by a factor 4 when we double the number of quadrature nodes (i.e. half the distance h between the quadrature nodes). This is the expectation for a method with second-order convergence.

This becomes even clearer if we plot the error versus h in a log-log plot along with a line of slope 2:



▶ **Why is quadratic convergence not surprising?**

Simpson's rule

```
# TODO("Show an illustrative drawing as well")
```

In the construction of the trapezoidal rule we employed linear approximations $\widetilde{f}_{1,i}$ of f in each of the subintervals $[\tau_{i-1}, \tau_i]$. One may thus wonder: Why stop at using only linear polynomials?

Indeed, **Simpson's formula** takes the idea one step further and constructs a **quadratic approximation** $\widetilde{f}_{2,i}$ within each interval $[\tau_{i-1}, \tau_i]$ by evaluating f on the interval boundaries τ_{i-1}, τ_i, but also the midpoint $m_i = \frac{\tau_{i-1}+\tau_i}{2}$ (recall we need *three points* to fit a quadratic polynomial). Again using a Lagrange basis we write

$$\widetilde{f}_{2,i}(x) = f(\tau_{i-1}) \underbrace{\frac{x - m_i}{\tau_{i-1} - m_i} \frac{x - \tau_i}{\tau_{i-1} - \tau_i}}_{\text{Lagrange function of } \tau_{i-1}} + f(m_i) \underbrace{\frac{x - \tau_{i-1}}{m_i - \tau_{i-1}} \frac{x - \tau_i}{m_i - \tau_i}}_{\text{Lagrange function of } m_i} + f(\tau_i) \underbrace{\frac{x - \tau_{i-1}}{\tau_i - \tau_{i-1}} \frac{x - \tau}{\tau_i - \tau}}_{\text{Lagrange funct. of}}$$

Employing this approximation in construction (2) gives

$$\int_a^b f(x)\,dx = \sum_{i=1}^{n} \int_{\tau_{i-1}}^{\tau_i} f(x)\,dx \approx \sum_{i=1}^{n} \int_{\tau_{i-1}}^{\tau_i} \widetilde{f}_{2,i}(x)\,dx$$

which requires us to integrate the quadratic polynomials $\widetilde{f}_{2,i}$ over the intervals $[\tau_{i-1}, \tau_i]$. This is a little harder to compute and will be done as an exercise. The resulting formula is Simpson's formula

$$\begin{aligned}
\int_a^b f(x)\,dx \approx S_{2n}(f) &= \sum_{i=1}^{n} \int_{\tau_{i-1}}^{\tau_i} \widetilde{f}_{2,i}(x)\,dx \\
&= \sum_{i=1}^{n} \frac{h}{6}\big(f(\tau_{i-1}) + 4f(m_i) + f(\tau_i)\big) \qquad \text{(S)} \\
&= \frac{h}{6} f(\tau_0) + \frac{h}{3}\sum_{i=1}^{n-1} f(\tau_i) + \frac{2h}{3}\sum_{i=1}^{n} f(m_i) + \frac{h}{6} f(\tau_n).
\end{aligned}$$

While a little harder to see, this formula can also be brought into the form of (3): it employs $N = 2n+1$ **equispaced nodes** — namely the collection of both the τ_i for $i = 0, \ldots, n$ *and* the m_i for $i = 1, \ldots n$:

$$\begin{aligned}
t_1 &= \tau_0 = a \\
t_2 &= m_1 = a + \frac{1}{2}h \\
t_3 &= \tau_1 = a + h \\
&\vdots \\
t_{2n-1} &= \tau_{n-1} = a + (n-1)\,h \\
t_{2n} &= m_n = a + \frac{2n-1}{2}h \\
t_{2n+1} &= \tau_n = b.
\end{aligned}$$

Therefore $N = 2n+1$ in (3) leading to a **nodal distance** of $\frac{b-a}{N-1} = \frac{b-a}{2n} = \frac{h}{2}$ while the **subinterval size** remains as $h = \tau_{i+1} - \tau_i = \frac{b-a}{n}$.

Exercise

Complete the missing step in the derivation of Simpson's rule, i.e. prove that:

$$\int_{\tau_{i-1}}^{\tau_i} \widetilde{f}_{2,i}(x)\,dx = \frac{h}{6}\Big(f(\tau_{i-1}) + 4f(m_{i-1}) + f(\tau_i)\Big)$$

A Julia implementation of Simpson's rule is given below:

```
simpson (generic function with 1 method)
```

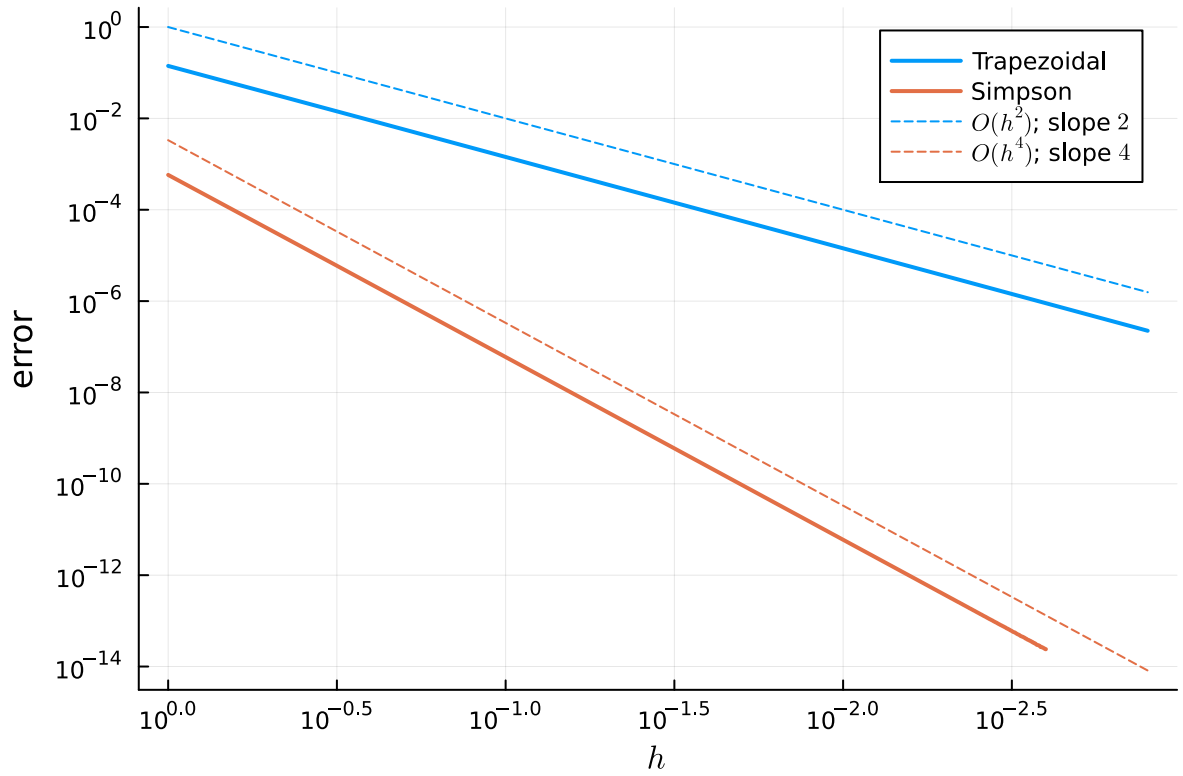
```julia
function simpson(f, a, b, n)
    # f:  Function
    # [a, b]: Interval to integrate over
    # n:  Number of subintervals
    h = (b - a) / n                     # Subinterval width
    τ = range(a, b, length=n+1)         # Subinterval boundaries
    m = range(a+h/2, b-h/2, length=n)   # Subinterval midpoints

    # Evaluate f and compute approximation
    fτ = f.(τ)
    fm = f.(m)
    integral = h * (fτ[1]/6 + sum(fτ[2:n])/3 + 2sum(fm)/3 + fτ[n+1]/6)

    (; integral, h)
end
```

Comparing the error of the trapezoidal and Simpson's quadratures against the exact integral of $\int e^x dx$ we obtain a much faster convergence for Simpson's rule, numerically looking like an $O(h^4)$ convergence.

```
p_convergence =
```




```
p_convergence = let
    # Trapezoidal rule
    Tfs     = [trapezoid(f, 0, 1, n).integral for n in 1:3:800]
    hs_trap = [trapezoid(f, 0, 1, n).h        for n in 1:3:800]
    p = plot(hs_trap, abs.(Tfs .- exact);
             yaxis=:log, xflip=true, xaxis=:log,
             xlabel=L"h", ylabel="error", label="Trapezoidal", lw=2)

    # Simpson's rule
    Sfs      = [simpson(f, 0, 1, n).integral for n in 1:2:400]
    hs_simps = [simpson(f, 0, 1, n).h        for n in 1:2:400]
    plot!(p, hs_simps, abs.(Sfs .- exact), label="Simpson", lw=2)

    # Guiding lines
    plot!(p, hs_trap, hs_trap.^2, ls=:dash, c=1, label=L"$O(h^2)$; slope $2$")
    plot!(p, hs_trap, hs_trap.^4/300, ls=:dash, c=2, label=L"$O(h^4)$; slope $4$")

    xticks!(p, 10.0 .^ (0:-0.5:-3))
    yticks!(p, 10.0 .^ (0:-2:-14))
end
```

In line with the definition of algebraic convergence (and convergence order) for other approximation techniques we define the accuracy of a quadrature formula as:

Definition: Convergence order of numerical integration

A numerical integration formula $Q_a^b(f)$ of the form (3) with equally spaced quadrature notes of separation h is of **order** p if a constant $\alpha > 0$ indepentent of h (but possibly dependent on f) exists, such that

$$\left| \int_a^b f(x)\,dx - Q_a^b(f) \right| \leq \alpha\, h^p$$

as long as the function f is sufficiently regular.

We notice that our numerical investigation suggests:

Convergence order of common numerical integration techniques

- **Trapezoidal rule:** Convergence order $p = 2$
- **Simpon's rule:** Convergence order $p = 4$

Error analysis

By construction composite quadrature formulas always satisfy

$$Q_a^b(f) = \sum_{i=1}^{n} Q_{\tau_{i-1}}^{\tau_i}(f)$$

where we use n subintervals in the numerical quadrature and $Q_{\tau_{i-1}}^{\tau_i}(f)$ is the application of the quadrature formula to exactly one such subinterval $[\tau_{i-1}, \tau_i]$. In other words $Q_{\tau_{i-1}}^{\tau_i}(f)$ is the application of the quadrature formula to $\int_{\tau_{i-1}}^{\tau_i} f(x)dx$ with no subdivision of the integration interval.

As a consequence we can decompose the error

$$\int_a^b f(x)\,dx - Q_a^b(f) = \sum_{i=1}^{n} \int_{\tau_{i-1}}^{\tau_i} f(x)\,dx - Q_{\tau_{i-1}}^{\tau_i}(f) \tag{4}$$

into error contributions from each of the subintervals $[\tau_{i-1}, \tau_i]$. Assume for simplicity that the function f is smooth and we can thus build a Taylor expansion

$$f(x) = \sum_{k=0}^{\infty} \frac{1}{k!} f^{(k)}(m_i)\,(x - m_i)^k$$

around the midpoint $m_i = \frac{\tau_i + \tau_{i-1}}{2}$ of the interval $[\tau_{i-1}, \tau_i]$. Based on this we can deduce a series for the exact integral:

$$\begin{aligned}
\int_{\tau_{i-1}}^{\tau_i} f(x)dx &= \int_{\tau_{i-1}}^{\tau_i} \left[\sum_{k=0}^{\infty} \frac{1}{k!} f^{(k)}(m_i)(x - m_i)^k \right] dx \\
&= \sum_{k=0}^{\infty} \frac{1}{k!} f^{(k)}(m_i) \int_{\tau_{i-1}}^{\tau_i} (x - m_i)^k dx \\
&= \sum_{k=0}^{\infty} \frac{1}{k!} f^{(k)}(m_i) \int_{\tau_{i-1}}^{\tau_i} q_k(x)\, dx,
\end{aligned}$$

where we defined $q_k(x) = (x - m_i)^k$ and in particular used that the integral is linear, i.e. that

$$\int_{\tau_{i-1}}^{\tau_i} \Big(\alpha f(x) + \beta g(x) \Big)\, dx = \alpha \int_{\tau_{i-1}}^{\tau_i} f\, dx + \beta \int_{\tau_{i-1}}^{\tau_i} g(x)\, dx.$$

Now notice that quadrature formulas of the form (3) keep this linearity in the integrand, i.e. also satisfy

$$Q_{\tau_{i-1}}^{\tau_i}(\alpha f + \beta g) = \alpha Q_{\tau_{i-1}}^{\tau_i}(f) + \beta Q_{\tau_{i-1}}^{\tau_i}(g).$$

Therefore we can make a similar development for the quadrature formula approximation of the ingeral, leading to

$$\begin{aligned}
Q_{\tau_{i-1}}^{\tau_i}(f) &= Q_{\tau_{i-1}}^{\tau_i} \left(\sum_{k=0}^{\infty} \frac{1}{k!} f^{(k)}(m_i)(x - m_i)^k \right) \\
&= \sum_{k=0}^{\infty} \frac{1}{k!} f^{(k)}(m_i) Q_{\tau_{i-1}}^{\tau_i}(q_k).
\end{aligned}$$

The difference between these expressions is exactly the error contribution from the interval $[\tau_{i-1}, \tau_i]$, namely

$$\int_{\tau_{i-1}}^{\tau_i} f(x)\, dx - Q_{\tau_{i-1}}^{\tau_i}(f) = \sum_{k=0}^{\infty} \frac{1}{k!} f^{(k)}(m) \left[\int_{\tau_{i-1}}^{\tau_i} q_k(x)\, dx - Q_{\tau_{i-1}}^{\tau_i}(q_k) \right]. \tag{5}$$

The **error of the integration formula** can thus be completely understood by studying the error $\int_{\tau_{i-1}}^{\tau_i} q_k(x)\,dx - Q_{\tau_{i-1}}^{\tau_i}(q_k)$. Since q_k is just a polynomial of degree k we can thus understand the accuracy of quadrature formulas *for arbitrary functions* by studying the **accuracy of quadrature formulas for polynomials**, which is considerably simpler.

One property of quadrature formulas is their **degree of exactness**:

Definition: Degree of exactness

A numerical integration formula $Q_a^b(f) = \frac{b-a}{N}\sum_{i=1}^n w_i f(t_i)$ as given in (3) has a **degree of exactness** r if it integrates all monomials x^s with $0 \le s \le r$ (s integer) exactly, i.e. if

$$Q_a^b(x^s) = \int_a^b x^s\,dx \qquad \forall s \in \mathbb{N} \text{ with } 0 \le s \le r, \tag{6}$$

but not for $s = r+1$.

Note that the polynomial

$$q_k(x) = (x-m_i)^k = x^k + \tbinom{k}{1} x^{k-1} m_i + \tbinom{k}{2} x^{k-2} m_i^2 + \cdots + \tbinom{k}{k-1} x\, m_i^{k-1} + m_i^k$$

only features monomials x^s with $0 \le s \le k$. Therefore a formula with degree of exactness r will have $\int_{\tau_{i-1}}^{\tau_i} q_k(x)\,dx - Q_{\tau_{i-1}}^{\tau_i}(q_k) = 0$ for $k \le r$. In (5) the first non-zero error term is thus

$$\begin{aligned}\left|\int_{\tau_{i-1}}^{\tau_i} q_{r+1}(x)\,dx - Q_{\tau_{i-1}}^{\tau_i}(q_{r+1})\right| &\overset{(*)}{=} \left|\int_{\tau_{i-1}}^{\tau_i} x^{r+1}\,dx - Q_{\tau_{i-1}}^{\tau_i}(x^{r+1})\right| \\ &\overset{(\S)}{\le} \tilde{\alpha}_i h^{r+2}\end{aligned}$$

where in $(*)$ all powers in x less than $r+1$ drop again because of Q's degree of exactness and in $(\S)$ we skipped a few non-trivial steps, which are optional and will be presented below. This is also the leading-order error term, such that

$$\left|\int_{\tau_{i-1}}^{\tau_i} f(x)\,dx - Q_{\tau_{i-1}}^{\tau_i}(f)\right| \le \tilde{\alpha}_i h^{r+2}$$

The error in each of the the n subintervals thus converges with $(r+2)$-th order, such that combining with (4) and using the triangle inequality we obtain the total error as

$$
\begin{aligned}
\left| \int_a^b f(x)\,dx - Q_a^b(f) \right| &\leq \sum_{i=1}^{n} \left| \int_{\tau_{i-1}}^{\tau_i} f(x)\,dx - Q_{\tau_{i-1}}^{\tau_i}(f) \right| \\
&\leq h^{r+2} \underbrace{\sum_{i=1}^{n} \widetilde{\alpha}_i}_{n \text{ terms}} \\
&\leq h^{r+1} \frac{b-a}{n}\, n \max_i \widetilde{\alpha}_i \\
&= \alpha\, h^{r+1}
\end{aligned}
$$

where $\alpha = (b-a)\ \max_i \widetilde{\alpha}_i$.

We notice:

Theorem 1 (Convergence order, simple version)

A composite quadrature formula with **degree of exactness r is of order $r+1$** as long as the function f is at least $r+1$ times differentiable.

▶ **Optional: More details on Theorem 1**

In short by **determining the degree of exactness** of a quadrature formula we thus **automatically obtain its convergence order**.

In fact in agreement with our expected convergence orders (2 and 4) one can show that:

Degree of exactness of common numerical integration techniques

- **Trapezoidal rule:** Degree of exactness $r = 1$
- **Simpson's rule:** Degree of exactness $r = 3$

We will show this now explicitly for the case of Simpson's rule.

Example: Simpson's formula has degree of exactness r=3.

We will confirm that Simpson's formula has degree of exactness $r = 3$.

Recall that the idea of Simpson's formula was to split the integral into n equally sized subintervals $[\tau_{i-1}, \tau_i]$ as

$$\int_a^b f(x)\,dx = \sum_{i=1}^{n} \int_{t_{\tau-1}}^{\tau_i} f(x)\,dx$$

and then on such a subinterval approximate

$$\int_{\tau_{i-1}}^{\tau_i} f(x)\,dx \approx \frac{h}{6}\big(f(\tau_{i-1}) + 4f(m_i) + f(\tau_i)\big). \tag{7}$$

If the approximation in (7) is an equality on all subintervals for the monomials with degree ≤ 3, then Simpson's formula is an equality for these monomials on the full interval $[a, b]$ as well. Inserting $f(x) = x^s$ into (7) we notice that our task is to show

$$\int_{\tau_{i-1}}^{\tau_i} x^s\,dx = \frac{h}{6}\left(\tau_{i-1}^s + 4m_i^s + \tau_i^s\right) \qquad \text{for } 0 \leq s \leq 3,$$

but that this does not hold for $s = 4$.

An additional trick we can employ to simplify our calculations is to assume that the integration interval is $[\tau_{i-1}, \tau_i] = [-1, 1]$ with a length of $h = 2$, $\tau_{i-1} = -1$, $\tau_i = 1$, $m_i = 0$. While perhaps suprising this does not actually change the generality of our result [1]. Inserting the values for h, τ_{i-1}, τ_i and m_i into the above expression we need to show that $0 \leq s \leq 3$

$$\int_{-1}^{1} x^s\,dx = \frac{2}{6}\left((-1)^s + 4\cdot 0^s + (+1)^s\right) = \frac{(-1)^s + 4\cdot 0^s + (+1)^s}{3},$$

but that this equality fails for $s \geq 4$.

We have

$$\begin{aligned}
s = 0: &\qquad \int_{-1}^{1} 1\,dx = 2 = \frac{(-1)^0 + 4\cdot 0^0 + 1^0}{3} && \checkmark \\
s = 1: &\qquad \int_{-1}^{1} x\,dx = 0 = \frac{-1 + 0 + 1}{3} && \checkmark \\
s = 2: &\qquad \int_{-1}^{1} x^2\,dx = \frac{2}{3} = \frac{(-1)^2 + 4\cdot 0^2 + 1^2}{3} && \checkmark \\
s = 3: &\qquad \int_{-1}^{1} x^3\,dx = 0 = \frac{(-1)^3 + 4\cdot 0^3 + 1^3}{3} && \checkmark
\end{aligned}$$

but in contrast

$$s = 4: \qquad \int_{-1}^{1} x^4\,dx = \frac{2}{5} \neq \frac{(-1)^4 + 4\cdot 0^4 + 1^4}{3} = \frac{2}{3}$$

Therefore Simpson's formula has **degree of exactness** $r = 3$.

[1]:

> The reason is that by an appropriate change of variables we can always rescale the integration coordinate, such that integration does run over $[-1, 1]$. Note that this in principle does generate some extra terms on top of x^s, but these are always of a lower power in x than x^s itself. So if we proceed as shown here, namely to consider integrating monomials with increasing powers of x, then all terms generated by this change of variables are already known to be integrated exactly at this stage.

Using other polynomial basis sets

In the construction of the Trapezoidal and Simpson's rules we decided to use equispaced nodes within the subintervals $[\tau_{i-1}, \tau_i]$ for building up our polynomial approximations $\widetilde{f}_i$ of the function f within this integral. This is of course just *one possible* choice.

In fact from our discussion about Interpolation techniques we only found equispaced polynomials to be suboptimal. Here are some alternatives:

- If we use **Chebyshev nodes** instead of equispaced nodes than we obtain **Clenshaw-Curtis integration**. See chapter 9.6 of Driscoll, Brown: Fundamentals of Numerical Computation for more details.
- Closely related is the idea to use an interpolation of f based on **Legendre polynomials**, resulting in **Gauss-Legendre integration**, see again chapter 9.6 of Driscoll, Brown for detaausils. This is one of the current state of the art numerical integration techniques. You will learn more about Gauss-Legendre integration in the exercises of this week.

Optional: Extrapolation techniques

```
# TODO("What is confusing here is that before we did expansions (I - T_n(f) = stuff, but now we do I = T_n(f) + stuff. )")
```

In numerical integration the **computationally expensive step** is usually the **evaluation of the function** f at the points of the quadrature nodes. Generally higher-order numerical integration formulas are able to achieve higher accuracy with the same or less function evaluations. We will discuss one systematic technique to obtain higher-order quadrature formulas called **extrapolation**.

Suppose a quantity A_0 is approximated by an algorithm A_h with error expansion

$$A_0 = A_h + c_1 h + c_2 h^2 + c_3 h^3 + \cdots = A_h + c_1 h + c_2 h^2 + O(h^3). \tag{8}$$

Crucially it is not necessary to know the coefficients c_i, the only requirement is for them to be indepentent of h. If we now look at the better approximation $A_{h/2}$ with half the node spacing then

$$A_0 = A_{h/2} + \frac{c_1}{2} h + \frac{c_2}{4} h^2 + O(h^3) \tag{9}$$

Forming a **linear combination $2 \cdot (9) - (8)$ between both methods** we obtain

$$A_0 = \underbrace{2A_{h/2} - A_h}_{=\widetilde{A}_{h/2}} - \frac{c_2}{2} h^2 + O(h^3)$$

which **defines a new approximation algorithm $\widetilde{A}_{h/2} = 2A_{h/2} - A_h$**. Note that while A_h is a first-order algorithm, $\tilde{A}_{h/2}$ is a **second-order algorithm**.

This idea to cancel the leading-order error term by taking a linear combination of two formulas of simple and half node spacing is termed **Richardson extrapolation**. Importantly in contrast to simply reducing the node spacing this scheme is able to **increase the convergence order**.

Let us **apply this to the trapezoidal formula** for approximating the integral $I = \int_a^b f(x)\,dx$. We use $n+1$ quadrature nodes of equal separation $h = (b-a)/n$. As we have discussed above the trapezoidal formula is of order 2, so the leading-order error term is h^2. However, in this fortunate case one can even show (using the Euler–Maclaurin formula) that the odd powers of h are missing, i.e.

$$I = T_n(f) + c_2 h^2 + c_4 h^4 + O(h^6), \tag{10}$$

where

$$c_2 = -\frac{1}{12}\big(f'(b) - f'(a)\big) \qquad c_4 = \frac{1}{740}\big(f'''(b) - f'''(a)\big).$$

While it could happen that $f'(b) = f'(a)$ we will make the general assumption that $c_2 \neq 0$.

For convenience of notation we rewrite (10) in terms of the number of subintervals. Using $n = O(h^{-1})$ we obtain

$$I = T_n(f) + c_2\, n^{-2} + c_4\, n^{-4} + O(n^{-6}). \tag{11}$$

The Trapezoidal rule with twice the number of subintervals (half the spacing) similarly reads

$$I = T_{2n}(f) + \frac{1}{4} c_2 \, n^{-2} + \frac{1}{16} c_4 \, n^{-4} + O(n^{-6}). \tag{12}$$

Using an appropriate linear combination we can again cancel out the second-order term. Specifically, define

$$S_{2n}(f) = \frac{1}{3}\Big(4\, T_{2n}(f) - T_n(f)\Big) \tag{13}$$

and consider $\frac{4}{3} \cdot (12) - \frac{1}{3}(11)$:

$$I = S_{2n}(f) - \frac{1}{4} c_4 \, n^{-4} + O(n^{-6}). \tag{14}$$

The quadrature formula $S_{2n}(f)$ is thus a 4-th order method. While not immediately clear (14) turns out to be identical to Simpson's formula (3), see the details at the end of this section. In other words we can view **Simpson's formula** as the numerical integration formula obtained either

- by integrating a piecewise quadratic approximation of the integrand or
- by applying Richardson extrapolation to the trapezoidal rule.

We formulate a general procedure for Richardson extrapolation in the context of numerical integration:

Observation: General procedure for Richardson extrapolation

Given an error expansion of a quadrature Q_n with n subintervals

$$I = Q_n + c_p n^{-p} + O(n^{-q}), \tag{15}$$

where p is integer and $q > p$. Then a a **quadrature of at least order q** can be found by the linear combination

$$\widetilde{Q}_{2n} = \frac{2^p Q_{2n} - Q_n}{2^p - 1}. \tag{16}$$

This can be verified as follows: using twice the number of nodes in (15) we obtain

$$I = Q_{2n} + 2^{-p} c_p n^{-p} + O(n^{-q}). \tag{17}$$

Then considering the linear combination $\frac{2^p}{2^p-1}(17) - \frac{1}{2^p-1}(15)$ between the equations of single and double the number of subintervals we obtain

$$I = \widetilde{Q}_{2n} + O(n^{-q}),$$

i.e. that $\widetilde{Q}_{2n}$ is of order at least q.

Note that Simpson's formula (13) turns out to be another error expansion of the form (8), so **we can apply Richardson extrapolate another time**, this time using Simpson's rule with $2n$ and $4n$ quadrature subintervals. From (16) we note the appropriate linear combination to be

$$R_{4n}(f) = \frac{1}{15}\Big(16 S_{4n}(f) - S_{2n}(f)\Big), \tag{18}$$

which is a **6-th order numerical integration formula**.

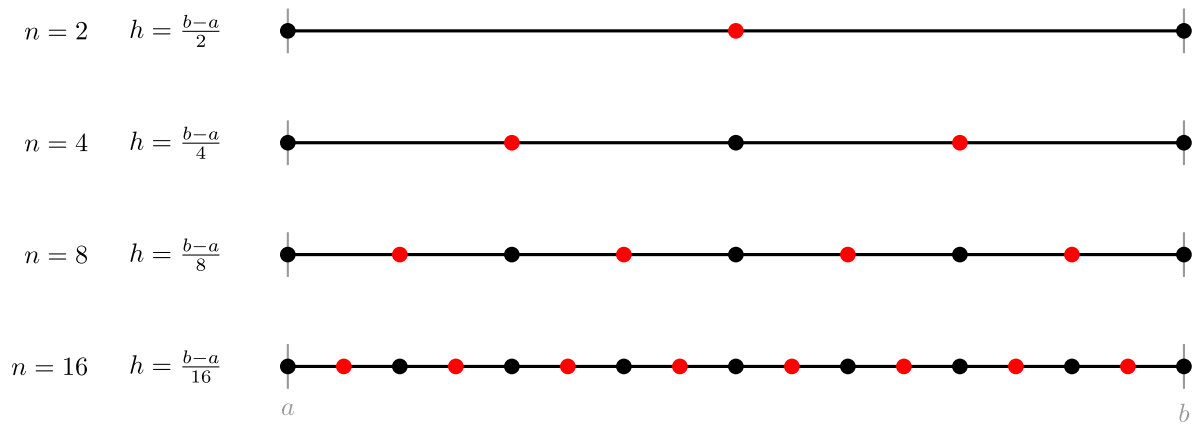
This process can of course be repeated using $8n$, $16n$, $\ldots$ subintervals, leading to higher and higher quadrature orders. One usually calls this **Romberg integration**, which we will not present in full generality here.

▶ **Optional: Details on the equality of the two forms of Simpson's formula**

Optional: Node doubling

If we consider the integration formula R_{4n} we notice that R_{4n} depends on S_{4n} and S_{2n}, which in turn depend on T_n, T_{2n} and T_{4n}. Continuing along the **Romberg integration procedure** we would **recursively perform more and more levels of extrapolation**. At the next level we also require T_{8n}, then T_{16n} and so on, such that **in each step** the **number of nodes** to consider roughly **doubles**.

This **node doubling is particularly advantageous in practice** and much preferred over other schemes. The reason is illustrated in the figure below, which divides the node spacing by half from one level to the next. In each layer new nodes are only introduces at the midpoints (shown in red). Notably, about half of the total number of nodes in each layer is red. Therefore moving from h to $h/2$ (respectively from $n+1$ to $2n+1$ nodes) **only about half of the nodes are new** and need to be evaluated. On the others the function values are already known (as they were needed at the prevous level) and can be re-used without extra cost.




```
# TODO("Using the drawing above this can be better explained, in particular T_{2n} easily re-written from T_n if one considers halfing the interval size ... perhaps some extra labels need to be introduced in the drawing above")
```

To make this explicit we use a formulation for the trapezoidal rule, which is developed in the part *Details on the equality of the two forms of Simpson's formula*, equation (19). As it states the trapezoidal rule with $2n$ subintervals can be expressed in terms of quantities of the trapezoidal rule with n subintervals, that is in terms of the quadrature nodes $\tau_i = a + i\,h$ (for $i = 0, \ldots, n$), the nodal spacing $h = (b-a)/n$ and the midpoints $m_i = \frac{\tau_{i-1}+\tau_i}{2}$:

$$T_{2n}(f) = \frac{h}{4} f(\tau_0) + \frac{h}{2} \sum_{i=1}^{n} f(m_i) + \frac{h}{2} \sum_{i=1}^{n-1} f(\tau_i) + \frac{h}{4} f(\tau_n).$$

Noting $m_i = a + (2i-1)\,\frac{h}{2}$ for $i = 1, \ldots, n$ and using the definition of $T_n(f)$ this can be reformulated as

$$T_{2n}(f) = \frac{1}{2} T_n(f) + \frac{h}{2} \sum_{i=1}^{n} f\left(a + (2i-1)\,\frac{h}{2}\right). \tag{20}$$

Since $a + (2i-1)\,\frac{h}{2} = a + (2i-1)\,\frac{b-a}{2n}$ for $i = 1, \ldots, n$ are just the *odd* quadrature nodes of the $T_{2n}(f)$ quadrature formula, we see that only these n odd nodes need to be evaluated additionally when considering $T_{2n}(f)$ instead of $T_n(f)$.

Let's see this in an example. We want to compute

$$V = \int_0^2 x^2 e^{-2x}\, dx$$

using extrapolation. First we use `quadgk` to get an accurate value:

```julia
begin
    g(x) = x^2 * exp(-2x)
    a = 0
    b = 2
    V, _ = quadgk(g, a, b, atol=1e-14, rtol=1e-14)
end;
```

```
0.1904741736116139
```

```julia
V
```

Based on the trapezoidal rule on $n = 20$ subintervals we get a first estimate:

```
T20 = 0.19041144993926787
```

```julia
T20 = let
    n = 20
    h = (b - a) / n
    t = h * (0:n)
    y = g.(t)
    T20 = h*y[1]/2 + h*sum(y[2:n]) + h*y[n+1]/2
end
```

Now we double to $n = 40$, but we only need to evaluate the odd nodes:

```
T40 = 0.19045880585951175
```

```julia
T40 = let
    n = 40
    h = (b - a) / n
    t = h * (0:n)
    y_odd = g.(t[2:2:n])
    T40 = T20/2 + h*sum(y_odd)
end
```

We repeat a second time and double n again:

```
T80 = 0.1904703513046443
```

```julia
T80 = let
    n = 80
    h = (b - a) / n
    t = h * (0:n)
    y_odd = g.(t[2:2:n])
    T80 = T40/2 + h*sum(y_odd)
end
```

Using (13) we can perform a first level of extrapolation and get the two Simpson values $S_{40}(f)$ and $S_{80}(f)$:

```
S40 = 0.19047459116625973
```

```
S40 = (4T40 - T20) / 3
```

```
S80 = 0.19047419978635513
```

```
S80 = (4T80 - T40) / 3
```

Finally, we perform one more level of extrapolation to get the sxth-order accurate result $R_{80}(f)$:

```
R80 = 0.1904741736943615
```

```
R80 = (16S80 - S40) / 15
```

We compute all errors to 10 digits:

```
begin
    eT = round.(V .- [T20 T40 T80]; digits=10)
    eS = round.(V .- [S40 S80];     digits=10)
    eR = round.(V .- [R80];         digits=10)
end;
```

and summarise them in a table along with the number of function evaluations:

number of evals	order 2	order 4	order 6
21	6.27237e-5		
41	1.53678e-5	-4.176e-7	
81	3.8223e-6	-2.62e-8	-1.0e-10

We notice that the 6th order **result obtained using Richardson extrapolation** is about **twice as accurate** as the order 2 result, even though it uses the **same number of function evaluations**.

Since the cost of function evaluation is usually dominating in numerical integration, the take-away is that one should **never just use low-order quadratures**, but always **employ extrapolation techniques.**

Optional: *A posteriori* error estimation

If we want to numerically evaluate an integral $I = \int_a^b f(x)\,dx$ using a chosen quadrature formula Q_n, a natural question to ask is how many quadrature nodes n are necessary to obtain an approximation of the the integral I within a predefined error tolerance ϵ.

In the previous sections we discussed, that Richardson extrapolation provides a receipe to obtain a higher-order quadrature $\widetilde{Q}_{2n}(f)$ from the numerical integration values $Q_n(f)$ and $Q_{2n}(f)$. As a

result $\widetilde{Q}_{2n}(f)$ is in general more accurate than $Q_{2n}(f)$, which motivates to employ the difference

$$\eta_{2n} = |\widetilde{Q}_{2n}(f) - Q_{2n}(f)|$$

as an estimate of the error committed by the formula $Q_{2n}(f)$. Indeed assuming an error expansion (15), i.e. $I = Q_n(f) + c_p n^{-p} + O(n^{-q})$, and and a corresponding Richardson extrapolation (16) one verifies

$$\begin{aligned} |I - Q_{2n}(f)| \leq |I - \widetilde{Q}_{2n}(f)| + |\widetilde{Q}_{2n}(f) - Q_{2n}(f)| &= \underbrace{\eta_{2n}}_{=O(n^{-p})} + \underbrace{|I - \widetilde{Q}_{2n}(f)|}_{=O(n^{-q})} \\ &= \eta_{2n} + O(n^{-q}) \end{aligned}$$

Therefore a simple adaptive strategy is to start with $2n$ nodes by computing and $Q_n(f)$ and $Q_{2n}(f)$. Then check whether $\eta_{2n} \leq \epsilon$ is satisfied and if this is not the case keep doubling the number of nodes until it is. Sticking to a single layer of extrapolation for simplicity we obtain the following algorithm:

Algorithm: Adaptive quadrature based on Richardson extrapolation

Given the problem to compute $\int_a^b f(x)\,dx$, quadrature formula $Q_n(f)$ an initial number of subintervals n_0 and a requested tolerance ϵ, compute:

- Initialise $n = n_0$, $h = (b-a)/n_0$
- Initial estimate of integral $I_n = Q_n(f)$
- While $\eta_n > \epsilon$ iterate:
  - Update $h = \frac{b-a}{2n}$
  - $I_{2n} = Q_{2n}(f)$ (using updated node distance h)
  - Extrapolate $\tilde{I}_{2n} = \frac{2^p I_{2n} - I_n}{2^p - 1}$ using (14)
  - $\eta_{2n} = |\tilde{I}_{2n} - I_{2n}|$
  - Update $n = 2n$

Applied to the trapezoidal formula this is implemented as follows:

```
trapezoid_adaptive (generic function with 1 method)
```

```julia
function trapezoid_adaptive(f, a, b; n₀=2, tol=1e-12)
    # f:  Function
    # [a, b]: Interval to integrate over
    # n₀: Initial number of subintervals (i.e. n₀+1 quadrature nodes)
    # tol: Desired tolerance
    n = n₀
    h = (b - a) / n₀  # Node separation
    t = (0:n) .* h   # Quadrature nodes
    y = f.(t)        # Evaluate function on all nodes
    Tₙ = h * (0.5y[1] + sum(y[2:n]) + 0.5y[n+1])  # Trapezoidal formula with n₀ nodes

    extrapolated = 0.0
    ηₙ = 10tol
    while ηₙ > tol
        h = (b - a) / 2n  # New node separation
        t = (0:2n) .* h   # New set of quadrature nodes
        y_odd = f.(t[2:2:2n])  # Evaluate only at odd nodes

        # Use node doubling formula (20) to evaluate Trapezoidal formula with 2n nodes.
        T₂ₙ = Tₙ/2 + h * sum(y_odd)

        # Perform extrapolation: Note that p = 2 for the Trapezoidal formula
        extrapolated = (4T₂ₙ - Tₙ) / 3

        n  = 2n
        Tₙ = T₂ₙ
        ηₙ = abs(extrapolated - T₂ₙ)
    end

    (; integral=extrapolated, h, n)
end
```

We test this works as expected using our example from the node doubling section:

```
▸ (integral = 0.190474, h = 0.03125, n = 64)
```

```julia
(; integral, h, n) = trapezoid_adaptive(g, 0, 2; tol=1e-5)
```

The error is indeed below the requested tolerance:

```
6.386761788879092e-8
```

```julia
abs(integral - V)
```

The next step to improve this approach would be to additionally employ Romberg integration, i.e. to recursive extrapolate not only from $T_{2n}(f)$ & $T_{4n}(f)$ to $S_{4n}(f)$, but to also employ $T_n(f)$ and

evaluate the 6-th order formula $R_{4n}(f)$ like we did in the numerical example of the node doubling section. However, these ideas are out of scope for us.

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